

Melt-pool geometry and solidification time in IN625 laser melting: sensitivity to high-temperature property extrapolation

Abstract

Melt-pool dimensions provide accessible calibration targets for thermal models of laser additive manufacturing, yet the solidification history inferred from a fitted field also depends on its thermophysical description. For IN625, the property tables used in the present model stop below the melting interval, leaving the resulting rear temperature profile dependent on extrapolation. Here, an enthalpy-based conduction model of bare-plate laser tracks resolves that dependence through paired rear solidus and liquidus positions. Conductivity and sensible specific heat are extrapolated using either the final tabulated slope or a constant endpoint value, separately and together, while retaining the source-power fraction fitted to one condition's thermographic mean length. The length fit leaves the fusion envelope wider and shallower than the metallographic means. Constant endpoint extrapolation of conductivity lengthens the pool, while the corresponding specific-heat change shortens it. The dominant length increase displaces both rear phase boundaries downstream with much less change in their separation. Across two scan speeds, almost equal separations nevertheless correspond to a passage time through the adopted freezing interval shorter by about one-third at the higher speed. These results connect prescribed property functions to distinct spatial and temporal features of the calibrated field. They identify rear phase-boundary position, separation and material passage time as specific outputs for thermal assessment alongside fusion geometry.

Keywords: Laser powder bed fusion, IN625, Melt-pool geometry, Enthalpy model, High-temperature properties, Model discrepancy

1. Introduction

Thermal models of laser powder bed fusion (LPBF) connect processing conditions to the extent of melting and to the thermal conditions experienced during solidification. Melt-pool geometry identifies the fused region, whereas solidification analysis also uses temperature gradients and material cooling histories. Heat-transfer calculations have been used to relate process inputs to these thermal quantities and to assess grain-structure trends in selected laser and electron-beam powder-bed cases [1]. Spatially and temporally resolved thermography provides complementary observations of temperature profiles and cooling during laser processing [2]. Geometry and temperature history thus serve related but distinct purposes: a geometric fit constrains a melting boundary, while the thermal information associated with that boundary depends on the model used to calculate it.

Moving-source conduction models provide this information without resolving the full motion of the liquid or the absorbing surface. Distributed-source formulations remove the near-source singularity of concentrated-source solutions, and nonlinear enthalpy formulations accommodate temperature-dependent properties and latent heat [3]. Their practical value is illustrated by the IN625 benchmark conducted on the Additive Manufacturing Metrology Testbed (AMMT) [4]. Using the measured beam profile, Kollmannsberger et al. fitted a scalar absorbed-power parameter that reproduced length more closely than width and depth; a calibrated directional-conductivity model then improved the geometric agreement across the tested conditions [5]. This improvement depended on the material

description as well as the source. Their conductivity data ended at 871 °C, below the melting range, and they explicitly treated its high-temperature extrapolation as a physical model requiring calibration [5]. The resulting geometry therefore reflects both the incident heating and the assumed transport beyond the available material data.

The thermal information in such a calibrated field requires a separate assessment. The AMMT modeling study restricted the use of its improved model for predicting temperatures inside the melt pool [5]. A later study of bare-plate 316L tracks demonstrated why temperature observations add information to a geometric fit: several Fresnel and accommodation coefficient pairs in a fluid model gave comparable agreement with cross-sectional area and width-to-depth ratio, but predicted different surface-temperature profiles. Two-color measurements distinguished those thermal predictions [6]. This evidence establishes the importance of the particular observable used to assess a model. In the AMMT measurements, length is obtained from a radiance profile, while width and depth are obtained from metallographic cross sections [4]. The dimensions constrain different features of the calculated field, and their agreement does not select the material functions that generated its temperature distribution.

The IN625 material functions used in the present study make this dependence concrete. Conductivity determines diffusive heat transport, while sensible specific heat enters the enthalpy stored during heating and released during cooling. The supplier tables provide conductivity only to 982 °C and sensible specific heat only to 1093 °C, both below the listed 1290–1350 °C melting range. The bulletin describes conductivity as measured and specific heat as calculated typical alloy data [7]. Both functions must therefore be extended before the model can evaluate transport and sensible storage throughout the adopted melting interval and above it. Assessments of nickel-superalloy properties also show that liquid-property estimates depend on the measurements and estimation procedure available [8]. The fitted source-power factor is consequently conditional on the chosen extrapolation functions; a geometric calibration does not supply the missing high-temperature material data.

The extrapolation of sensible properties must be distinguished from the phase-change prescription in the same enthalpy balance. Latent heat is stored and released according to the adopted liquid-fraction relation as well as the total heat of fusion [3]. A linear fraction across a specified freezing interval is an established continuum approximation [9]. Its endpoints, however, need not coincide across material descriptions: an IN625 model for rapid additive-manufacturing cooling used a lower eutectic endpoint from Gulliver–Scheil calculations, whereas the supplier bulletin lists a melting range [9, 7]. Changing that interval redistributes latent storage along the temperature axis. Comparing conductivity and sensible-heat-capacity extrapolations with a common phase law therefore gives their responses a defined meaning within the same thermal model.

With the phase thresholds fixed, the rear temperature profile can resolve features that total pool length combines. A solidus-crossing length depends on the front and rear boundary positions. At the rear, the solidus and liquidus crossings describe both the position of the thermal tail relative to the source and the spatial separation across the adopted freezing interval. Displacing the two rear crossings together changes total length while preserving approximately the same separation. Changing their separation alters the spatial interval over which material cools between the adopted thresholds. Conductivity and sensible storage affect the same coupled field, so an overall length response alone cannot distinguish these changes. Resolving the rear pair provides a direct way to interpret which part of the calculated thermal structure responds to the prescribed property functions.

The material time associated with this spatial interval also depends on scan speed. For a steadily translating field, material fixed in the laboratory passes through the rear solidus–liquidus separation in a time equal to that separation divided by the speed. Similar spatial intervals can thus correspond to different passage times and interval-mean cooling rates. Constant-speed conversion from position to time is used in the AMMT cooling analysis, although that analysis considers a below-solidus

interval [4]. The trajectory represented by the conversion remains consequential: pulsed LPBF thermography resolves histories affected by successive heating events [2], and fluid calculations show that liquid motion changes the thermal excursions followed by material parcels [10]. Passage through a quasi-steady conduction field describes a stationary-material trajectory whose time interpretation is tied to that field and its adopted freezing interval.

Taken together, the need to extrapolate the material inputs and the different information carried by geometry and material time motivate the present assessment of a length-calibrated IN625 conduction field. In a nonlinear moving-frame enthalpy model, an effective source-power fraction is fitted to condition B thermographic length and retained for two complementary comparisons across three AMMT bare-plate conditions. Width, depth and the other operating conditions assess the geometry associated with that fit. At B, independent conductivity and sensible-heat-capacity contrasts compare linear extrapolation using the final tabulated slope with constant endpoint extrapolation under a common phase law. Resolving the rear solidus–liquidus positions and separation distinguishes thermal-tail displacement from a change in interval width, while the operating-condition comparison relates each spatial interval to material passage time at its scan speed. The objective is to determine how the prescribed property functions alter the spatial and temporal interpretation of the calibrated field.

2. Materials and methods

The analysis uses two comparisons to examine the thermal information in a length-calibrated field. The first retains the B-fitted effective source-power fraction when assessing width, depth and the other operating conditions. The second changes the above-table conductivity and sensible specific heat at B, with B’s source power and the phase law retained. Rear solidus and liquidus positions distinguish displacement of the cooling interval from a change in its spatial extent; scan speed then converts that extent to a passage time for stationary material.

2.1. Benchmark observations and comparison design

The benchmark comprises constant-speed single-track laser scans on bare IN625 plate made on the Additive Manufacturing Metrology Testbed (AMMT) [4], under the three public conditions in Table 1. Conditions B and C share the same applied power and differ in scan speed; condition A combines a lower power with a lower speed. The effective source-power fraction η scales the power supplied to the conduction model to ηP . It is fitted to B length and retained in the comparisons described below. The measured beam diameter is $D_{4\sigma} = 170 \mu\text{m}$ [4]. For the assumed circular Gaussian, $D_{4\sigma} = 2w$, giving $w = 85 \mu\text{m}$. This reconstruction matches the measured second-moment diameter with circular symmetry; the published beam measurement retains a two-dimensional intensity map.

Table 1: AMMT operating conditions and model energy scales. The effective source-power fraction $\eta \approx 0.28905$ is fitted to B thermographic length only. $E_\ell = \eta P/v$ and w/v are model input scales.

Condition	P (W)	v (m/s)	E_ℓ (J/m)	w/v (μs)
A	137.9	0.4	99.65	212.50
B	179.2	0.8	64.75	106.25
C	179.2	1.2	43.17	70.83

Length targets are the AMMT-20 μs thermographic class means reported by Lane et al. [4]. Their A/B/C populations contain 19/10/7 video frames, respectively. Width and depth are compared with the NIST challenge Table 2 metallographic means for the AMMT-100 μs specimen: three tracks each

for A and B, and four for C [11]. Lane’s later cross-section compilation combines the 100 and 20 μs microscopy classes [4]. Length and cross-sectional dimensions consequently enter as condition-level measurements from separate sampling populations, rather than three jointly observed dimensions of one pool.

The published geometric targets were consulted during model development, so the post-calibration comparisons are retrospective and nonblind. Only condition B length enters the calibration objective; B width/depth and all A/C dimensions are assessed with the resulting source-power fraction. Expanded uncertainties U with coverage factor $k = 2$ are taken from Lane’s uncertainty budgets [4]. For width and depth, these scales belong to the combined-class microscopy analysis, whereas the target means belong to the NIST 100 μs classes. Each signed difference $\delta = Q_{\text{model}} - Q_{\text{measurement}}$ is also reported as δ/U , a descriptive normalization by that published measurement scale. The denominator does not include model, input or calibration uncertainty, and the width/depth ratios do not constitute a formal uncertainty test of the 100 μs target means.

2.2. Moving-frame enthalpy model and high-temperature properties

The conduction model accounts for sensible and latent energy through a conservative enthalpy formulation [3]. This permits conductivity and sensible specific heat to be changed separately while preserving one phase law. The laboratory-frame balance is

$$\frac{\partial H}{\partial t} = \nabla \cdot [k(T)\nabla T], \quad H(T) = \rho \left[\int_{T_0}^T c_p(s) ds + \mathcal{L}f(T) \right]. \quad (1)$$

The density $\rho = 8440 \text{ kg m}^{-3}$ follows the IN625 supplier bulletin; its listed melting range supplies the adopted thresholds $T_s = 1290 \text{ }^\circ\text{C}$ and $T_l = 1350 \text{ }^\circ\text{C}$ [7]. The latent heat $\mathcal{L} = 280000 \text{ J kg}^{-1}$ follows the earlier AMMT model [5]; the reference temperature is set to $T_0 = 25 \text{ }^\circ\text{C}$. A linear liquid-fraction approximation distributes this latent heat over the adopted interval. A linear phase-fraction relation is also used in a continuum IN625 model [9], although that study selects a different freezing interval. Here the liquid fraction is

$$f(T) = \begin{cases} 0, & T < T_s, \\ (T - T_s)/(T_l - T_s), & T_s \leq T < T_l, \\ 1, & T \geq T_l. \end{cases} \quad (2)$$

Sensible specific heat and conductivity are linearly interpolated between the input knots in Table A.4. The conductivity table contains 15 points and ends at 982 $^\circ\text{C}$; the 12-point specific-heat table ends at 1093 $^\circ\text{C}$. The supplier labels specific heats as calculated and conductivity as measured, with the conductivity measurements made at Battelle on material annealed at 2100 $^\circ\text{F}$ for 1 h [7]. These are typical alloy data rather than benchmark-coupon measurements. Within the adopted phase interval, differentiating Eq. (1) gives $c_{p,\text{eff}} = c_p + \mathcal{L}/(T_l - T_s)$, with a latent contribution of 4666.67 $\text{J kg}^{-1} \text{ K}^{-1}$. Retaining this contribution in every case distinguishes a change in sensible specific heat from a change in latent heat.

Both tables end below the adopted solidus, so the solution through the melting interval requires an above-table property definition. The closest AMMT conduction study likewise identified high-temperature conductivity extrapolation as a modeling assumption that participates in calibration [5]. Figure 1 defines the two extrapolation rules examined here.

For either property $p \in \{k, c_p\}$, let (T_{n-1}, p_{n-1}) and (T_n, p_n) be its final two tabulated pairs. Above T_n , final-secant linear extrapolation (L) and constant endpoint extrapolation (H) are defined by

$$p_L(T) = p_n + \frac{p_n - p_{n-1}}{T_n - T_{n-1}}(T - T_n), \quad p_H(T) = p_n, \quad T > T_n. \quad (3)$$

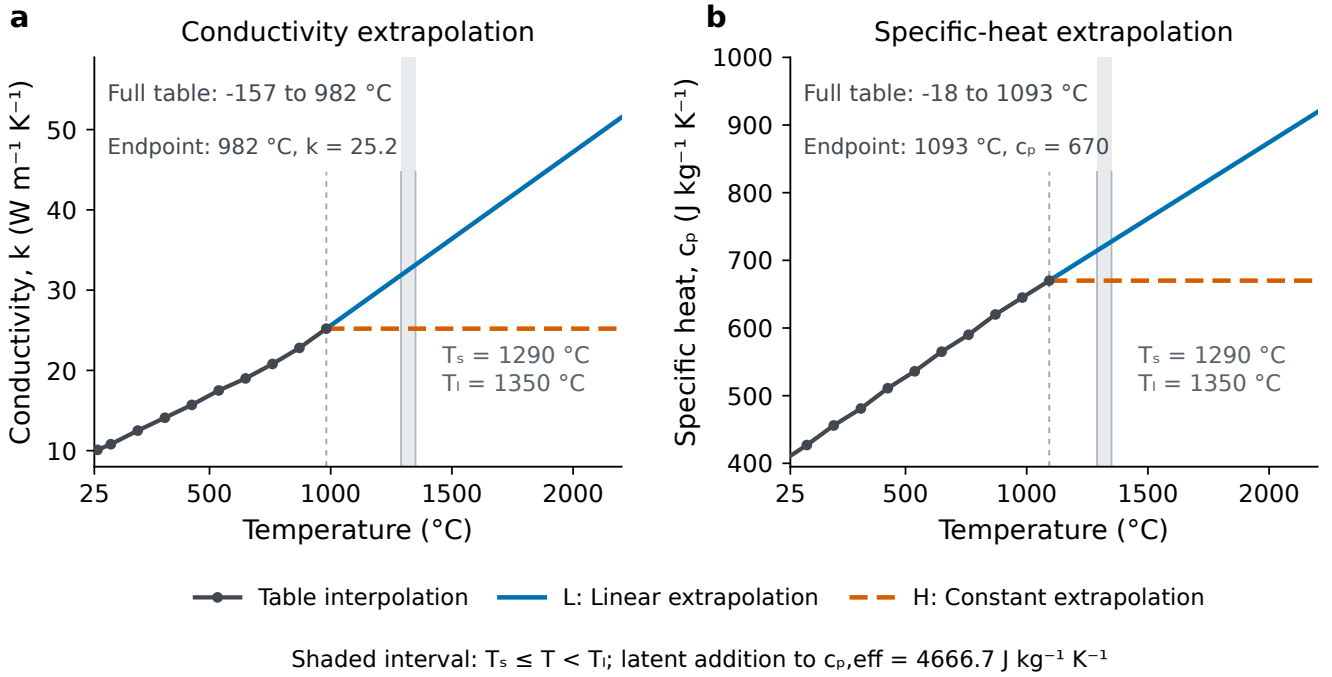


Figure 1: IN625 property inputs and extrapolation definitions. (a) Conductivity and (b) sensible specific heat use the knots in Table A.4, with common piecewise-linear interpolation below their respective endpoints. The final knots are 982 °C and 25.2 W m⁻¹ K⁻¹ for k , and 1093 °C and 670 J kg⁻¹ K⁻¹ for c_p . L denotes final-secant linear extrapolation and H constant endpoint extrapolation, as defined in Eq. (3). The displayed window is 25–2200 °C; all 15 conductivity and 12 specific-heat source rows, including those below 25 °C, are retained in the accompanying data and used in interpolation. Shading marks the adopted 1290–1350 °C phase interval. The latent addition to effective specific heat within this interval is 4666.7 J kg⁻¹ K⁻¹, separate from sensible c_p . The supplier describes conductivity as measured and specific heat as calculated typical alloy data [7]; the above-table functions are study-defined choices.

Thus H uses $k = 25.2 \text{ W m}^{-1} \text{ K}^{-1}$ above 982 °C and $c_p = 670 \text{ J kg}^{-1} \text{ K}^{-1}$ above 1093 °C; the appendix lists the final-secant slopes. Each change applies above its table endpoint, including the solid range between that endpoint and T_s . Conductivity changes are applied consistently to k , its primitive K and the inverse used for surface-temperature recovery. Specific-heat changes are applied to c_p , the enthalpy H and its inverse $T(H)$. The archived two-letter labels place conductivity first: LL uses linear extrapolation for both properties, HL and LH change one property, and HH changes both. These functions define numerical property scenarios rather than bounds on measured liquid properties.

Following the moving-source coordinate description [3], let $\xi = x - vt$. For a quasi-steady field, $\partial_t H = -v \partial_\xi H$; defining $\mathbf{u} = (-v, 0, 0)$ therefore gives

$$\nabla \cdot (\mathbf{u}H - k \nabla T) = 0. \quad (4)$$

The term $\mathbf{u}H$ represents stationary material passing through the laser coordinate frame. No liquid momentum equation or melt convection is included. The assumptions also exclude evaporation and free-surface deformation.

The circular Gaussian surface input at $z = 0$, normalized consistently with the $D_{4\sigma}$ convention [9], is

$$q(\xi, y) = \frac{2\eta P}{\pi w^2} \exp\left[-\frac{2(\xi^2 + y^2)}{w^2}\right]. \quad (5)$$

Its full-plane integral is ηP , so fitting η adjusts the total input power while retaining the source shape; it does not measure absorptivity independently of the adopted model. With outward normal

$\mathbf{n}$, the top condition is $-k\nabla T \cdot \mathbf{n} = -q$, supplying inward heat. The appendix gives the discrete source integration. At positive ξ , incoming material has $H = 0$; at negative ξ , diffusive gradient vanishes and enthalpy leaves advectively. The $y = 0$ plane is symmetric. Outer transverse and bottom boundaries are adiabatic. Radiation and ambient convection are omitted from the solved balance.

2.3. Geometry, rear phase boundaries and material passage

The model solidus-crossing length L is the distance between the front and rear crossings of the recovered surface symmetry-centerline temperature. These crossings are interpolated, must enclose the source, and must form one interval without domain truncation. Surface and symmetry-plane recovery precede threshold extraction; the appendix gives their numerical definitions. Retaining both positions allows a change in total length to be assigned to motion at the front or rear.

Fusion-zone width and depth are extracted from the source-connected component of

$$T_{\max}(y, z) = \max_{\xi} T(\xi, y, z) \geq T_s. \quad (6)$$

Each ξ slice is reconstructed at the surface and symmetry plane, bilinearly interpolated onto observation coordinates, and then maximized over passage. This order defines the maximum-temperature fusion envelope; maximizing nodal temperatures before interpolation gives a different envelope. Width is twice the positive- y extent and depth is the negative of the lowest z . The observation pitch controls sampling of the reconstructed geometry. Figures label the model surface-length proxy L_s and fusion-envelope extents W_f and D_f ; these correspond to L , W and D in the text. A separate surface-sample maximum is taken over the recovered top-surface temperatures at axial and positive- y cell-center coordinates, before the $y = 0$ extension. It therefore has a different sampling definition from the symmetry-centerline field used for the crossings.

Lane's thermographic length locates the rear freezing feature from a radiance-profile second-derivative minimum, then finds the front at the assumed freezing temperature after emittance-based conversion [4]. This measured length includes the effects of emittance, exposure and viewing geometry. The model instead uses two solidus crossings of a recovered temperature field. Its fitted η consequently belongs to the combination of source, material law and boundary definition used here. Fusion-envelope width and depth are compared with the metallographic dimensions defined in Section 2.1.

Let ξ_s^- and ξ_l^- denote the rear centerline solidus and liquidus crossings. Their positions locate the adopted freezing interval relative to the source; their separation gives its spatial extent. Define this rear solidus–liquidus separation, passage time and interval-mean cooling-rate magnitude as

$$\ell_m = \xi_l^- - \xi_s^-, \quad \tau_m = \frac{\ell_m}{v}, \quad \bar{T} = \frac{T_l - T_s}{\tau_m}. \quad (7)$$

At a fixed laboratory point, $d\xi/dt = -v$, so the material traverses the rear interval in ℓ_m/v . The same spatial separation can therefore correspond to different passage times at different scan speeds. Constant-speed space-to-time mapping is also used in the AMMT measurements [4], but their cooling estimate covers a below-solidus interval, while Eq. (7) uses 1350–1290 °C. Lane et al. present their cooling rates as exemplar data rather than model-calibration references [4]. Here, separation, time and interval-mean cooling rate are derived from the same modeled field; the passage follows stationary material rather than a moving liquid parcel.

2.4. Length calibration and property comparisons at retained source power

With LL properties, fitting $L_B(\eta) = 359 \mu\text{m}$ gives $\eta \approx 0.28905$, with a signed length residual of $+0.158 \mu\text{m}$. Conditions A and C retain this value, beam, phase law and property extrapolations.

The appendix gives the safeguarded scalar calibration algorithm, trial values and local sensitivity. This length fit sets the effective source-power fraction across A, B and C, and fixes the source power within the four B property cases.

The four B cases share power, speed, beam, density, phase thresholds, latent heat, mesh, observation operators and the source-power fraction fitted with LL properties. Only the above-table k and c_p definitions change, including their corresponding primitives and recovery operations. The first letter denotes conductivity and the second sensible specific heat. Sharing B=LL between the three-condition geometry comparison and the four-case property comparison gives six production solutions. Retaining the LL-fitted factor makes the other B cases responses at common source power, rather than separate length fits.

For any output Q , define the baseline-anchored finite effects

$$\delta_k = Q_{\text{HL}} - Q_{\text{LL}}, \quad \delta_{c_p} = Q_{\text{LH}} - Q_{\text{LL}}, \quad (8)$$

$$I_{k,c_p} = Q_{\text{HH}} - Q_{\text{HL}} - Q_{\text{LH}} + Q_{\text{LL}}, \quad \Delta_{\text{joint}} = Q_{\text{HH}} - Q_{\text{LL}} = \delta_k + \delta_{c_p} + I_{k,c_p}. \quad (9)$$

At constant-extrapolated c_p , the conductivity effect is $Q_{\text{HH}} - Q_{\text{LH}} = \delta_k + I_{k,c_p}$; at constant-extrapolated k , the specific-heat effect is $Q_{\text{HH}} - Q_{\text{HL}} = \delta_{c_p} + I_{k,c_p}$. Reporting these conditional effects establishes whether each response direction persists at both levels of the other property. The interaction records the departure of the joint change from the sum of the two LL-anchored changes. All effects are deterministic finite differences between specified functions within this conduction model.

An axial-face Péclet diagnostic compares coordinate enthalpy transport with diffusion in the phase interval. A discrete rear hot-region budget measures net outward diffusive power. Each branch defines its temperature-selected region separately, allowing the diagnostic population and region boundary to follow the solved field. These budgets complement the matched-coordinate phase thresholds; their definitions are given in the appendix.

2.5. Numerical solution and verification

The conservative finite-volume solution uses near-source cell widths of $4 \times 3 \times 1 \mu\text{m}$. Gaussian power is integrated over each top-face rectangle and supplied to its adjacent control volume; surface-temperature recovery uses the corresponding conductivity primitive. Conductivity and enthalpy relations are updated with the nonlinear solution, so each extrapolation case changes the material functions consistently. The graded-domain geometry, scaled enthalpy discretization and recovery equations are specified in the appendix. Convergence is checked using both the sum of absolute cell residuals and the signed global energy balance, followed by a full frozen-coefficient correction.

A separate constant-property moving-Gaussian problem checks source integration, frame transport and geometric extraction against a semi-infinite analytical reference. This verification isolates numerical implementation from assessment against physical measurements [12]. Its reference $L/W/D$ is 407.535/153.293/34.836 μm ; finite-volume differences are +0.139%, +0.190% and -0.499%, respectively. The benchmark uses constant properties and omits phase change, making its accuracy assessment specific to that verification problem.

At the calibrated η , reducing the near-source axial spacing from 4 to 2 μm changes B length, width and depth by +0.361, -0.009 and +0.041 μm , and C by +0.199, -0.029 and +0.045 μm . A/B domain enlargement changes these quantities by less than $3 \times 10^{-6} \mu\text{m}$. These measured sensitivities support the interpretation of the larger geometric responses, but do not give a full three-dimensional error bound for every extrapolation case. Submicrometre interactions are consequently reported descriptively.

An earlier transient B comparison at a different calibration stage provides approximate local steady consistency. Fixed-field blackbody estimates put B-case top losses at 0.099–0.180% of ab-

sorbed power; feedback from these losses is not solved. These checks have distinct roles from the constant-property verification and the calibrated-mesh comparisons.

All six solutions satisfy the stated energy-balance and nonlinear-correction criteria and give bounded phase fractions and nontruncated extracted geometry. The appendix reports the acceptance thresholds and final residuals.

3. Results

The post-calibration geometry comparison identifies which aspects of the fusion shape remain discrepant after fitting length. The B property comparison then resolves whether its length response arises mainly from motion of the rear phase boundaries or from a change in their separation. The fixed-power B–C comparison finally relates the spatial separation to material passage time at different scan speeds.

3.1. Geometry discrepancy after length calibration

Fitting the source-power fraction to B thermographic length gives a model length of 359.16 μm , but leaves a wider and shallower fusion envelope than the B metallographic means (Table 2 and Fig. 2). Model width exceeds the NIST 100 μs class mean by 19.54 μm , and depth falls below it by 4.88 μm . Normalizing these differences by Lane’s published combined-class microscopy uncertainty scales gives +3.10 and -1.23 , respectively. The fitted axial extent therefore does not reproduce the transverse fusion shape in this model.

Table 2: Model geometry and AMMT measurement means in μm . Length means are from Lane’s 20 μs thermography classes; width/depth means are from the current NIST 100 μs metallography classes. U is Lane’s separately sourced expanded measurement uncertainty ($k = 2$); width/depth U comes from its combined-class microscopy analysis. B length is fitted. δ/U is a descriptive difference normalized by that published scale, without combined model uncertainty.

Condition	Quantity	Model	Measured mean	U	δ/U
A	L	309.89	300	11.91	+0.83
	W	153.36	147.9	6.42	+0.85
	D	40.47	42.5	4.49	-0.45
B	L (fitted)	359.16	359	21.26	+0.01
	W	143.04	123.5	6.30	+3.10
	D	31.12	36	3.97	-1.23
C	L	320.63	370	32.57	-1.52
	W	129.05	106	4.48	+5.14
	D	22.04	29.6	3.26	-2.32

Condition A deviations are smaller than their respective published measurement-uncertainty scales. In C, the model is 49.37 μm shorter than the thermographic mean, 23.05 μm wider and 7.56 μm shallower than the NIST metallographic means. The modeled aspect ratios $W/D = 3.79, 4.60, 5.85$ for A/B/C also show the widening relative to penetration, compared with ratios of the measured class means of 3.48, 3.43, 3.58 (Fig. 5b). These measured ratios are formed from condition-level means, rather than paired with each thermographic length frame.

At fixed power from B to C, a 50% speed increase reduces model input energy per unit length, $\eta P/v$, by one third. Model width decreases by 9.78% and depth by 29.16%; the NIST metallographic class-mean changes are 14.17% and 17.78% (Fig. 5a). The modeled W/D consequently increases by 27.36% from B to C. The geometry discrepancy thus concerns both absolute dimensions and how

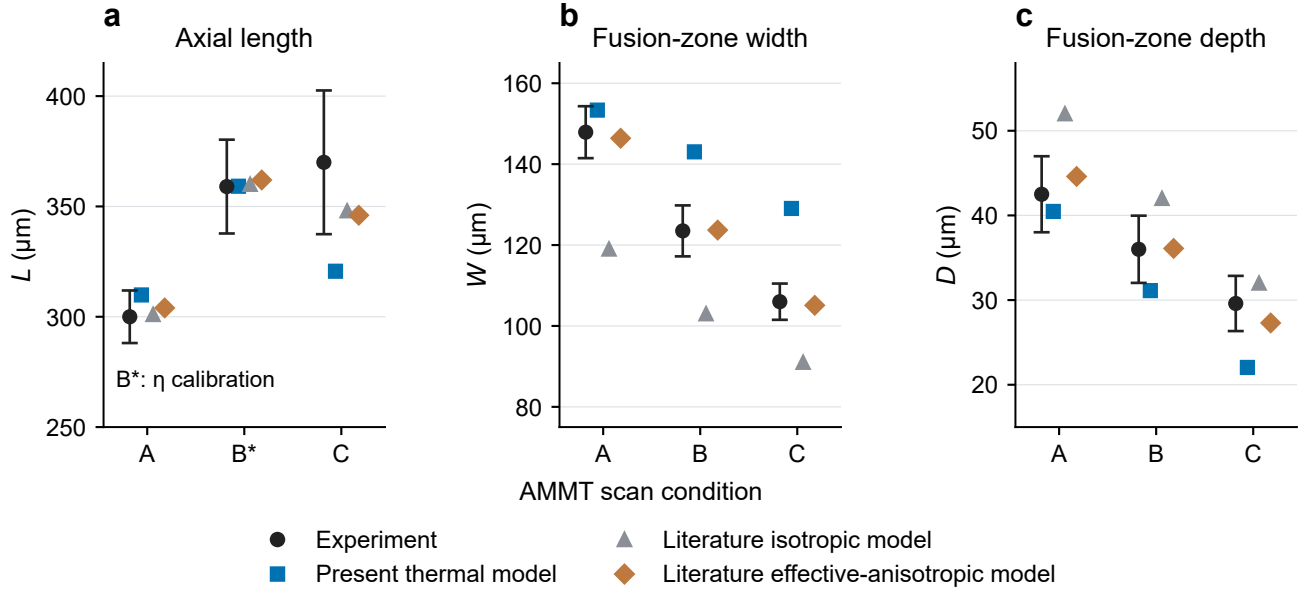


Figure 2: Geometry comparison with Lane’s 20 μs thermographic length means, NIST’s 100 μs metallographic width/depth means, and the published isotropic and effective-anisotropic calculations of Kollmannsberger et al. [5]. Experimental bars show separately sourced Lane $U(k = 2)$ scales; width/depth centers use current NIST means. B length is calibrated in the present model. Published-model points provide context under different beam representations, material/phase laws, calibration targets and observation definitions; their differences from the present results are not a single-factor comparison.

transverse contraction compares with reduced penetration as speed rises. Condition A also changes power, so the B–C pair provides the fixed-power speed comparison.

The thermographic class mean increases in length by 11 μm from B to C, whereas model length decreases by 38.53 μm . The two reported expanded-uncertainty scales give a zero-correlation quadrature of 38.89 μm , larger than the change between measured means. This descriptive scale comparison limits interpretation of the sign of the measured B–C trend; the C residual in Table 2 compares C with its own mean. The observed axial-refinement changes below 0.37 μm are much smaller than the main B/C residuals, so the retained refinement evidence does not remove the principal shape discrepancy.

The contours in Fig. 3 show the greater fractional reduction in penetration than in width for the fixed-power B–C comparison. The surface solidus contour supplies the model length, while the transverse envelope records the maximum melted extent during material passage. Since a length fit leaves this shape discrepancy, the B property comparison examines how the constitutive definitions change the fitted field’s rear boundaries, with the calibrated source-power fraction retained.

3.2. Property effects on rear-boundary position and separation

At the B-fitted source power, the two property changes produce opposing length responses (Table 3). Replacing final-secant linear extrapolation of conductivity by constant endpoint extrapolation increases length from 359.16 to 402.37 μm when c_p remains linear (LL to HL). Changing only c_p to constant endpoint extrapolation decreases length to 352.99 μm (LL to LH). Changing both yields 393.44 μm (HH), so

$$\Delta L_{\text{joint}} = 43.21 - 6.17 - 2.76 = 34.28 \mu\text{m}. \quad (10)$$

The specific-heat response and interaction offset part of the conductivity-induced increase. These response directions persist at the other factor level: constant extrapolation of conductivity increases

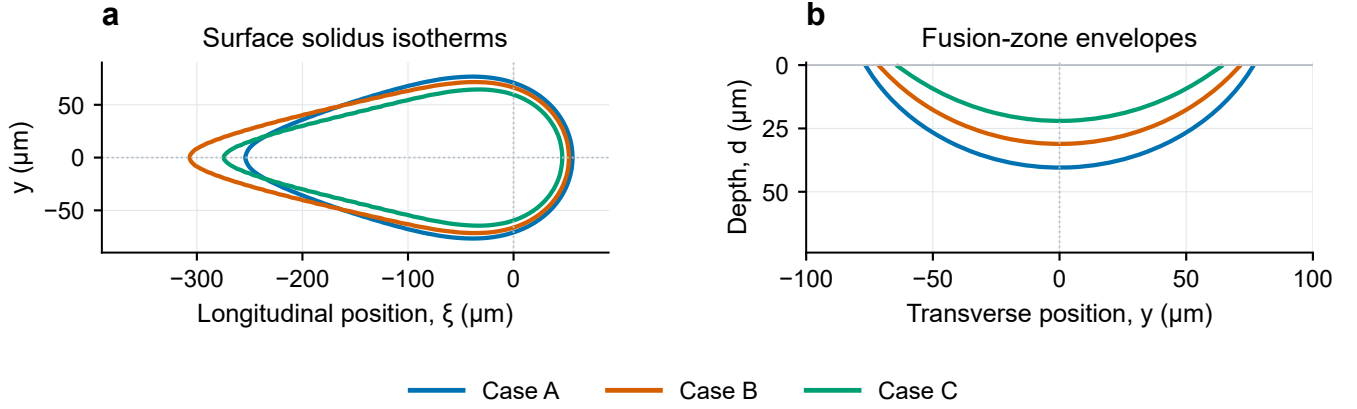


Figure 3: Model geometry for A, B and C. (a) Surface solidus contours in laser-frame coordinates. (b) Reconstructed maximum-temperature fusion envelopes in transverse/depth coordinates. Fusion envelopes use slice reconstruction and interpolation before passage maximization. Both panels display calculated thermal geometry and use the fixed calibrated source factor.

length by $40.45 \mu\text{m}$ at constant-extrapolated c_p , and constant extrapolation of c_p decreases length by $8.93 \mu\text{m}$ at constant-extrapolated k . Thus the direction of each length response is consistent across the two tested levels of the other property.

Table 3: Two-factor property comparison at B and the LL-fitted source-power fraction. The first letter labels k , the second c_p ; L denotes final-secant linear extrapolation and H constant endpoint extrapolation above the respective table endpoints. All cases retain the same source power and phase law. Rounded geometry and finite effects are in μm . Small transverse effects lack branch-specific refinement evidence.

Corner or contrast	L	W	D
LL	359.16	143.04	31.12
HL	402.37	143.15	30.09
LH	352.99	143.72	31.61
HH	393.44	143.79	30.80
δ_k	+43.21	+0.11	-1.03
δ_{c_p}	-6.17	+0.68	+0.49
I_{k,c_p}	-2.76	-0.04	+0.21
Δ_{joint}	+34.28	+0.75	-0.32

The rear solidus and liquidus positions identify which part of the field produces these length changes (Fig. 4). From LL to HL, the rear crossings move by -42.49 and $-43.50 \mu\text{m}$, respectively, while the front solidus moves by $+0.73 \mu\text{m}$. Rear solidus displacement accounts for 98.3% of the conductivity-induced length increase, with the front contributing 1.7%. Yet the rear solidus-liquidus separation changes by only $-1.01 \mu\text{m}$. From LL to LH, the rear crossings instead move towards the source by $+6.22$ and $+6.53 \mu\text{m}$, with a separation change of $+0.31 \mu\text{m}$. The large length response therefore arises predominantly from displacement of the pair of rear phase boundaries, rather than a comparable change in the distance between them.

The two prescribed changes perturb conductivity and total storage by different relative amounts. At 1320°C , changing L to H reduces conductivity from 32.51 to $25.20 \text{ W m}^{-1} \text{ K}^{-1}$, or 22.48%, and sensible specific heat from 721.13 to $670 \text{ J kg}^{-1} \text{ K}^{-1}$, or 7.09%. With the fixed latent contribution

included, effective specific heat decreases from 5387.79 to 5336.67 J kg⁻¹ K⁻¹, or 0.95%, and enthalpy from the cold reference decreases by 0.66%. Their geometric effect sizes consequently compare these specified input changes rather than equal-percentage perturbations of transport and total storage.

The auxiliary transport diagnostics also change from LL to HL. The median axial-face Péclet diagnostic rises from 4.44 to 5.77, while integrated outward diffusion from the rear hot region rises from 14.724 to 14.913 W. The latter increase accompanies the lower conductivity: the solved gradients and the temperature-selected region boundary both change. These values therefore describe each field's selected faces and region, rather than conductive power through one fixed integration boundary. The conductivity case also uses its corresponding material primitive in surface-temperature recovery.

The recovered positive- y surface-sample maxima defined in Section 2.3 are approximately 2697, 3507, 2756 and 3721 °C for LL, HL, LH and HH, respectively. Both cases with constant-extrapolated conductivity give higher sampled temperatures. These are maxima of the stated surface samples, not an independently resolved maximum temperature of the melt pool; they also differ in sampling from the recovered $y = 0$ profile used for phase-boundary positions. Their physical interpretation must retain the evaporation, flow and surface-deformation omissions of the conduction balance.

The current-mesh depth response contains opposing contributions. Constant extrapolation of conductivity changes depth by $-1.03 \mu\text{m}$ from LL; the specific-heat change contributes $+0.49 \mu\text{m}$ and the interaction $+0.21 \mu\text{m}$, leaving a joint change of $-0.32 \mu\text{m}$. Conductivity effects remain negative at both specific-heat levels (-1.03 and $-0.82 \mu\text{m}$), while specific-heat effects remain positive at both conductivity levels ($+0.49$ and $+0.71 \mu\text{m}$). Every case retains the B width excess and depth deficit. The extrapolation choices thus move the rear boundaries substantially while retaining the principal fusion-shape discrepancy. The 34–43 μm length response exceeds the retained baseline axial sensitivity by a wide margin; the 0.04 μm width interaction remains an unresolved fine-scale comparison pending branch-specific refinement.

Across the four cases, the rear solidus–liquidus separation lies between 79.25 and 81.17 μm , while total length ranges from 352.99 to 402.37 μm . The comparison therefore distinguishes a large response in the location of the rear freezing interval from a smaller response in its spatial extent. The fine ordering of those separations remains a current-mesh result.

3.3. Scan speed and derived material passage time

The fixed-power B–C comparison yields nearly equal rear solidus–liquidus separations at different scan speeds. On the present mesh, B and C give 80.26 and 80.79 μm , respectively, a difference of 0.53 μm (Fig. 5c). Their corresponding passage times are 100.33 and 67.33 μs (Fig. 5d). The C separation is 0.66% larger, yet its passage time is 32.90% shorter because speed increases from 0.8 to 1.2 m/s. The interval-mean cooling rate over 1350–1290 °C consequently rises from 0.598 to 0.891 MK s⁻¹, an increase of 49.02%. Condition A, which also changes power, gives a separation of 48.30 μm , a passage time of 120.74 μs and a cooling rate of 0.497 MK s⁻¹.

The 49.02% B–C cooling-rate increase follows from the shorter passage through an almost unchanged spatial interval, using Eq. (7). It is a second expression of the same centerline separation and scan speed. The spatial interval alone cannot determine the time experienced by material.

The B–C pair consequently shows both contraction of the fusion envelope and faster passage through an almost unchanged rear freezing interval. The former response is assessed against metallographic dimensions; the latter characterizes stationary material crossing the calculated centerline field. It complements the rear-position comparison by distinguishing where cooling occurs from the distance and time over which the adopted freezing interval is traversed.

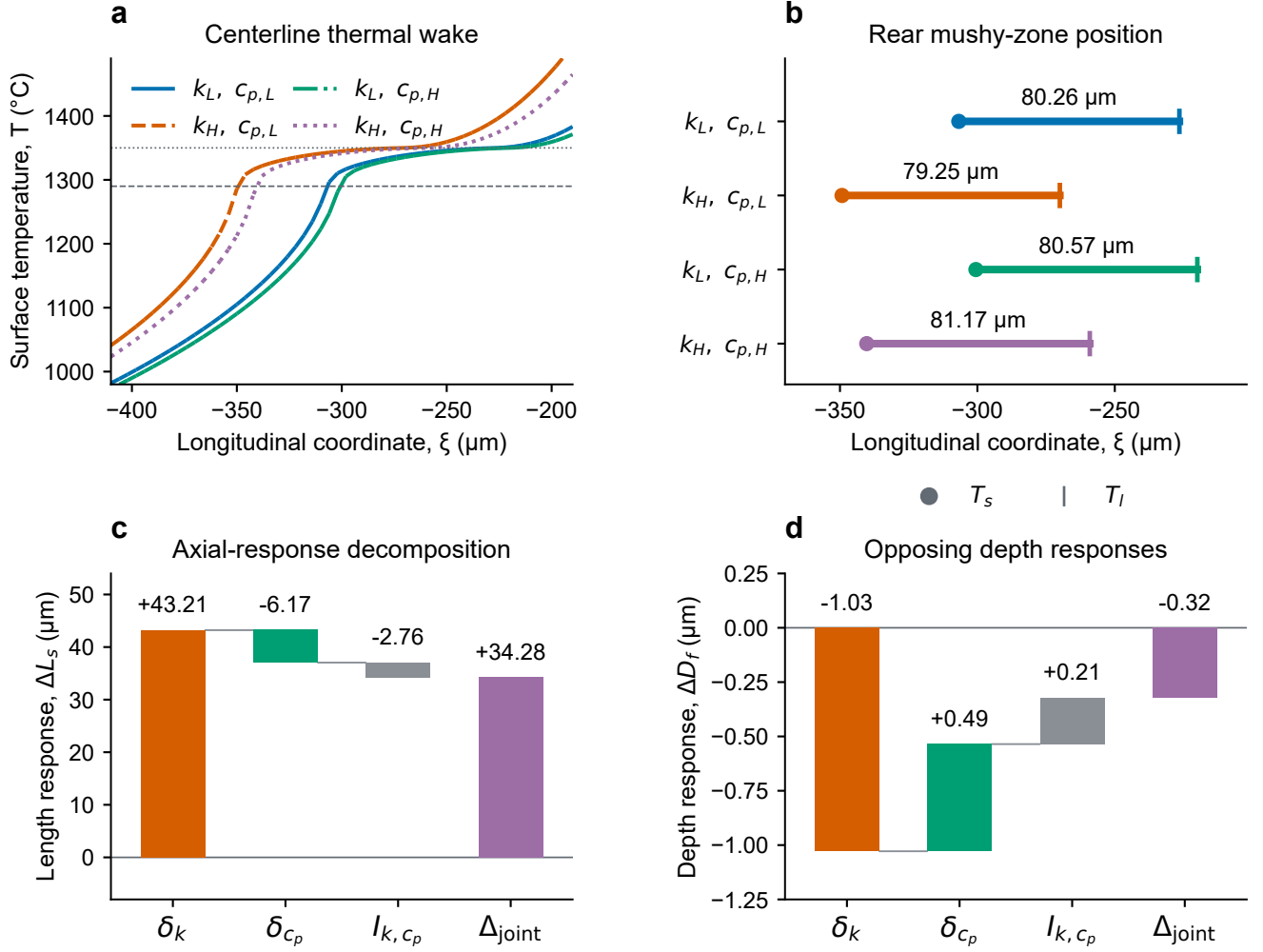


Figure 4: Property comparison at B: rear centerline temperature profiles, solidus/liquidus positions and their separation, and LL-anchored geometry effects. All four cases retain the same phase thresholds, calibrated source-power fraction and mesh. Profiles are plotted from original field values with segment clipping to the displayed window. Factorial effects are deterministic finite differences between the defined extrapolation functions. Displacement of the rear boundary pair dominates the length response; the joint depth change contains opposing single-property contributions.

4. Discussion

The geometric comparisons establish which dimensions remain inconsistent after the source-power fit. The property comparisons then locate the response within the rear temperature field, where boundary position, solidus–liquidus separation and material passage time describe different aspects of solidification. Their relationship determines both the interpretation of the calculated sensitivities and the measurements needed to assess them physically.

4.1. Constraints supplied by melt-pool geometry

Fitting the effective source-power fraction fixes the energy supplied by the adopted Gaussian to reproduce B’s thermographic mean length. The resulting wide, shallow fusion envelope shows that this axial constraint leaves the transverse dimensions unresolved. Width and depth add information about the distribution of melting across the track; their discrepancies cannot be removed by interpreting the length fit as agreement of the entire fusion boundary. The fitted fraction is conditional on

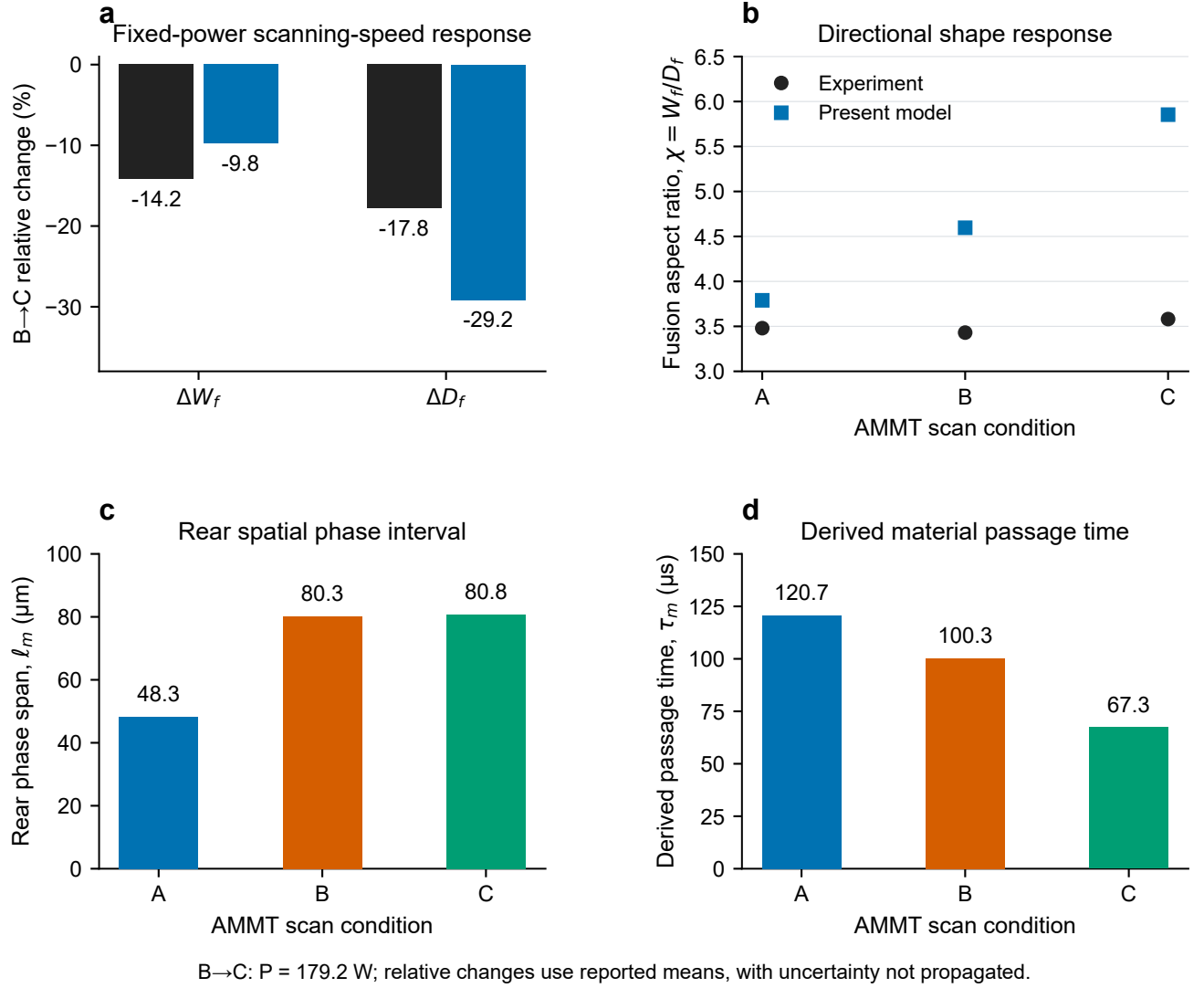


Figure 5: Operating-condition response. (a) B-to-C percentage changes in width and depth at fixed power, comparing the model with NIST 100 μs metallographic class means; uncertainty is not propagated. (b) Model and measured class-mean aspect ratios. (c) Model rear liquidus-to-solidus separations. (d) Derived material passage times. Separation and time are transformations of the same steady centerline field. The text reports the corresponding interval-mean cooling rates over 1350–1290 °C.

the Gaussian approximation, thermophysical functions, phase law and temperature-threshold length definition. Relating it to physical laser absorption requires an assessment of those choices and of the radiance-based length operator used by the experiment [4].

The closest AMMT study illustrates why these choices matter. With the measured beam profile, its isotropic calculation reproduced length more closely than width and depth; calibrated directional conductivity subsequently improved the dimensions [5]. Its isotropic shape was narrower and deeper than the reported metallographic means, whereas the present calculation is wider and shallower. The two calculations also differ in beam representation, property data and phase law. The predecessor identified the limits of fitting three dimensions with a single absorptivity parameter. Here, the transverse residuals identify the shape left unresolved by the B-length fit. The opposite residual directions prevent assigning the present discrepancy to a mechanism inferred from the predecessor. Within the present four-case B comparison, retaining the same source and phase law instead gives

the property-extrapolation differences a defined model-specific meaning.

Geometric constraints can also leave thermal predictions unresolved even when more source parameters are fitted. Myers et al. obtained similar cross-sectional area and width-to-depth ratio agreement for several Fresnel/accommodation parameter combinations in bare 316L calculations, but two-color temperature profiles distinguished those combinations [6]. Their study tests alternative fitted parameter sets. Here, the source fraction is retained after the baseline length fit, so the four IN625 cases measure the response to prescribed property functions. These are complementary designs: the former tests what geometric fitting leaves ambiguous, and the latter identifies how specified material assumptions change the fitted field.

4.2. Effects of property extrapolation on rear phase boundaries

Replacing linear extrapolation by constant endpoint values produces opposing responses of conductivity and sensible specific heat. Their combined length change is dominated by the conductivity contribution, whose rear solidus and liquidus displacements are almost equal. Resolving both positions shows that the longer pool principally reflects a downstream shift of the rear temperature profile. The much smaller change in their separation describes a different response: the spatial extent over which the adopted liquid fraction falls from one to zero. The persistence of the width excess and depth deficit across LL, HL, LH and HH further separates this axial response from correction of the transverse fusion shape.

The relative sizes of the imposed changes help interpret the unequal responses. Conservative enthalpy separates sensible storage from the adopted latent contribution [3]. At 1320 °C, constant endpoint extrapolation reduces conductivity by 22.48% and sensible specific heat by 7.09%. Because the phase law contributes 4666.67 J kg⁻¹ K⁻¹ to effective specific heat in this interval, effective specific heat decreases by only 0.95%. Integrating the modified specific-heat function from the cold reference also reduces enthalpy at this temperature by 0.66%. The conductivity and storage cases therefore apply unequal perturbations to diffusive transport and total thermal storage. Their effect sizes characterize these particular functions and the retained phase law. The specific-heat intervention also begins below the solidus, changing pre-melting enthalpy and the temperature–enthalpy inverse along with sensible storage in the freezing interval.

The conductivity response also involves the temperature gradients and the region over which conduction is evaluated. In the HL calculation, the median axial-face Péclet diagnostic increases and outward conductive power from the rear cells selected by $T \geq T_s$ and $\xi < 0$ rises from 14.724 to 14.913 W. Lower conductivity thus accompanies greater integrated outward power over this solution-dependent region. Both its boundary and its gradients change, so that integral cannot isolate reduced heat escape as the cause of the longer pool. In addition, consistent extrapolation changes the conductivity primitive and the recovered surface temperature used to locate the crossings. Comparing conductive power over an identical spatial region would remove the contribution from changing region selection. Separately evaluating surface-temperature recovery would quantify its contribution to the boundary displacement. Together, these diagnostics would narrow the interpretation of the complete calculation’s paired-boundary response.

4.3. Solidification interval in material coordinates

Rear-boundary displacement and separation acquire different time meanings in a steadily translating field. A common downstream displacement lengthens the pool without proportionally enlarging the interval between its rear liquidus and solidus. At a fixed speed, nearly unchanged separation consequently gives nearly unchanged passage time through the adopted freezing interval. Between B and C, the speed changes as well as the field: their separations differ by only 0.53 μm , but the faster C scan gives a 32.90% shorter passage time. The 49.02% increase in interval-mean cooling rate

follows from the same separation-and-speed relation. These descriptors distinguish the location of solidification relative to the laser from its duration for stationary material traversing the calculated field.

This distinction is consequential when comparing spatial thermography with a temperature history. Hooper measured centerline profiles and fixed-location histories that resolved heating pulses, cooling and turn events [2]. Lane et al. used a constant-speed position-to-time conversion for a below-solidus cooling interval [4]. The AMMT analysis uses a temperature range and radiance operator that differ from the present liquidus-to-solidus calculation. A direct thermal comparison would therefore require a measured trajectory aligned with the adopted interval, scan position and observation response. The existing passage times describe each condition’s own modeled field.

Melt motion changes the trajectories relevant to a microstructure calculation. Fluid-tracer calculations for a Ni–20Cr dispersion-strengthened alloy produced narrower near-surface temperature peaks and hotter subsurface excursions than the corresponding conduction histories [10]. Transient conduction calculations have also related solidification conditions to grain-structure trends in AlSi10Mg laser and IN718 electron-beam cases, with experimental grain-morphology comparison for the IN718 case [1]. These studies establish the importance of the thermal path and interface motion used in a microstructure model. The present straight-track, quasi-steady mapping supplies a stationary-material centerline interval; applying it to liquid-parcel histories, overlapping scans or grain evolution would require the corresponding transport and transient descriptions.

4.4. Material data and thermal measurements for model assessment

The numerical extrapolations make the sensitivity explicit, while their physical selection remains a material-data question. The supplier tables stop below the adopted melting interval, and neither defines a measured liquid-state plateau [7]. Composition-based nickel-superalloy estimates include IN625 conductivity and identify limited underlying liquid-state data [8]. Such estimates or composition- and state-matched measurements provide candidates for replacing the two prescribed extrapolation rules with physically informed functions. Their comparison should retain an explicit phase law: the dynamic-source IN625 study, for example, uses a 1410 K eutectic endpoint and phase-dependent properties that differ from the present supplier-listed interval [9].

Source deposition remains a separate candidate for the unresolved shape response. The dynamic volumetric model adjusts deposition depth and effective absorption with the calculated pool, and was calibrated on bare IN625 tracks across several beam diameters before application to a layer case [9]. Recent conduction-model assessment also uses the absorption required to reproduce penetration as an attainability diagnostic under specified transport assumptions [13]. Aligning source input, phase law, material functions and observation definitions is necessary before differences between these approaches can be assigned to source redistribution or transport. Combined thermography and metallography further show that geometry correlations can coexist with systematic errors outside stable conduction melting [14]; the operating regime belongs in that comparison.

The present extreme temperatures make this scope particularly consequential. LL, HL, LH and HH positive- y surface-sample peaks reach approximately 2697, 3507, 2756 and 3721 °C under a balance that omits evaporation, melt flow and surface deformation. These values expose sensitivity to the assumed heat balance and extrapolations; independent thermal measurements are needed to assess their physical credibility. The large 34–43 μm length responses exceed the inspected baseline axial sensitivity, whereas submicrometre interactions retain a need for branch-specific three-dimensional refinement. Analytical verification and residual closure support numerical execution within the inspected scope; physical assessment additionally depends on material data and the distinct experimental populations described in Materials and methods.

The resulting assessment has specific observable targets. Transverse metallography tests the fusion shape beyond the fitted length. Rear thermal profiles test the locations and separation of the adopted phase boundaries. A co-registered temperature history spanning that interval tests the derived material time. Keeping these roles distinct connects the constitutive sensitivity to evidence that can strengthen the calibrated model, rather than treating every geometric or derived thermal descriptor as an interchangeable validation target.

5. Conclusions

High-temperature property extrapolation is a consequential assumption in length-calibrated IN625 heat-conduction modeling because the available tables stop below the adopted melting interval. The present comparison makes its consequences explicit for fusion geometry, rear phase-boundary position and the passage time through that interval. Fitting one effective source-power fraction to B thermographic length leaves the calculated fusion envelope wider and shallower than the NIST metallographic means. The other dimensions therefore remain necessary constraints on the spatial distribution of melting.

At retained source power, constant endpoint extrapolation of conductivity lengthens the pool, while the corresponding specific-heat extrapolation shortens it; the combined increase is $34.28 \mu\text{m}$. Almost equal rear solidus and liquidus displacements account for the dominant conductivity response. Total length primarily changes through movement of this boundary pair, with much less change in its separation and with the transverse discrepancy retained. This distinguishes sensitivity of the rear phase-boundary location from an enlarged solidification interval or an improved fusion shape.

The interval’s time interpretation also requires scan speed. Nearly equal B/C rear separations at 0.8 and 1.2 m/s correspond to a roughly one-third shorter passage time and a 49.02% higher interval-mean cooling rate in C. Together, the results separate the spatial extent of melting, the location and separation of the rear phase boundaries, and the duration of cooling through the adopted interval. Material-specific high-temperature properties and aligned thermal observations can assess these distinct outputs of the calibrated field. The present study supplies their model-based comparison with one retained calibrated source-power fraction and a fixed phase law.

Data and code availability

The LaTeX manuscript, vector figures, extracted tables and evidence inspection are available in the [manuscript package](#). The original case and numerical implementation are referenced through [SimAgent](#). Experimental sources remain the cited NIST challenge and Lane study. Publisher full-text copies used for reading are not redistributed. The package README distinguishes extracted tables and figures from large retained fields and identifies the corresponding original case locators.

Declaration of generative AI use in manuscript preparation

OpenAI Codex was used to assist literature and evidence organization, manuscript drafting and automated pre-review. Figures were generated from numerical results by deterministic plotting.

Appendix A. Material data and numerical verification

Appendix A.1. Material-property inputs

The linear-extrapolation slopes from the final tabulated pairs are $0.0216216 \text{ W m}^{-1} \text{ K}^{-2}$ for conductivity and $0.225225 \text{ J kg}^{-1} \text{ K}^{-2}$ for sensible specific heat. The enthalpy integral is evaluated piecewise analytically and inverted using the corresponding property extrapolation.

Table A.4: Thermophysical-property inputs from the general IN625 bulletin [7]. The two temperature columns describe different tables. Conductivity and sensible specific heat are interpolated independently.

T_k (°C)	k (W m ⁻¹ K ⁻¹)	T_c (°C)	c_p (J kg ⁻¹ K ⁻¹)
-157	7.2	-18	402
-129	7.5	21	410
-73	8.4	93	427
-18	9.2	204	456
21	9.8	316	481
38	10.1	427	511
93	10.8	538	536
204	12.5	649	565
316	14.1	760	590
427	15.7	871	620
538	17.5	982	645
649	19.0	1093	670
760	20.8		
871	22.8		
982	25.2		

Appendix A.2. Spatial discretization and observation recovery

The domain is $\xi \in [-1, 0.35]$ mm, $y \in [0, 0.36]$ mm and $z \in [-0.30, 0]$ mm. A graded orthogonal mesh contains 902664 cells. Near-source cell widths are $4 \times 3 \times 1$ μm within $\xi \in [-0.48, 0.16]$ mm, $y \in [0, 0.12]$ mm and $z \in [-0.06, 0]$ mm. Outer widths increase by a factor of 1.22 up to 25 μm . The half-domain mesh has $199 \times 56 \times 81$ cells. The solved problem is steady; retained warm fields provide nonlinear initial guesses.

The Gaussian source is integrated rectangle-wise using error functions to supply each top-face power, which is deposited in the adjacent top control volume. Its full-plane integral is ηP and the transverse half-domain receives $\eta P/2$.

Surface recovery satisfies

$$K(T_{\text{surf}}) - K(T_{\text{cell}}) = q \Delta z/2, \quad K'(T) = k(T), \quad (\text{A.1})$$

and the symmetry-plane value uses the even quadratic extension $T(0) = T(y_1) + [T(y_1) - T(y_2)]y_1^2/(y_2^2 - y_1^2)$ from the first two positive- y cell centers. The recovered surface temperatures and symmetry extension are applied before the threshold crossings are extracted.

The observation pitch is 0.5 μm ; it samples the interpolated geometry rather than changing the heat-field resolution.

Appendix A.3. Scaled enthalpy diffusion and diagnostic definitions

The scaled unknown is $U = H/C_0$, with $C_0 = \rho(402 \text{ J kg}^{-1} \text{ K}^{-1}) = 3.39288 \times 10^6 \text{ J m}^{-3} \text{ K}^{-1}$. On an internal face,

$$D_f = \frac{k_f}{C_0} \frac{\Delta T}{\Delta U}, \quad (\text{A.2})$$

where k_f is the harmonic face conductivity and D_f has units of $\text{m}^2 \text{s}^{-1}$. For $|\Delta U| \leq 10^{-8} \text{ K}$, the derivative limit is $\Delta T/\Delta U = C_0/[\rho(c_p + \mathcal{L}f')]$ at the first adjacent cell. This coefficient recovers conductive flux in the conservative equation for scaled enthalpy U . Coordinate enthalpy transport

uses an exponentially fitted finite-volume convection flux. Material coefficients and the diffusion–transport stencil are updated with each nonlinear iterate; Anderson acceleration solves the fixed-point residual and preconditioned GMRES solves the inner linear system. Software versions and iterative controls are given in the accompanying numerical reproduction notes.

The retained face transport diagnostic is

$$|\text{Pe}_f| = \frac{|\mathbf{u} \cdot \mathbf{n}_f| d_f}{D_f}, \quad (\text{A.3})$$

where d_f is the adjacent cell-center separation. Unweighted medians are taken over internal axial faces adjacent to at least one cell with $T_s \leq T < T_l$. The diagnostic compares coordinate transport with diffusion in the discretized field; its selected face population can change between solved branches.

The discrete rear hot-region budget selects cells with $T \geq T_s$ and $\xi < 0$. Net outward diffusive power is the sum of signed conductive face powers on the boundary of this selected cell set; internal face contributions cancel. The region is selected separately in each field, so its boundary and sampled faces change with the property extrapolation. This accounting diagnostic is interpreted alongside the matched-coordinate centerline thresholds rather than as a fixed-region causal contrast.

Appendix A.4. Scalar calibration and solution acceptance

A safeguarded secant solves $L_B(\eta) - 359 \mu\text{m} = 0$ in the initial bracket $[0.2, 0.4]$, with proposals restricted to the inner 90% of the current bracket and an accepted absolute length residual below $0.5 \mu\text{m}$. It yields $\eta = 0.2890548519203547$, with a residual of $+0.158 \mu\text{m}$. The calibration record below gives trial values and the local sensitivity conversion. A and C retain the final value, the same beam and phase law, and the same extrapolation rules.

Nonlinear acceptance requires the sum of absolute assembled cell residuals, normalized by absorbed half-power, to be below 2×10^{-5} , the signed global balance magnitude below the same scale, and a further frozen-coefficient solve to change temperature by less than $0.002 \text{ }^\circ\text{C}$. Local residual and global balance are retained separately because opposing local errors can cancel in a global sum.

The six production solutions have finite temperatures, bounded phase fractions, nontruncated extracted geometry and the prescribed integrated source power. Normalized absolute residuals are 6.90×10^{-8} – 1.31×10^{-7} , signed global balance magnitudes are below 6.1×10^{-11} , and full-solve temperature corrections are below $2.7 \times 10^{-5} \text{ }^\circ\text{C}$.

Appendix A.5. Separate constant-property verification problem

The Gaussian verification uses $P = 179.2 \text{ W}$, $\eta = 0.2$, $v = 0.8 \text{ m/s}$ and $w = 85 \mu\text{m}$, with $\rho = 8440 \text{ kg m}^{-3}$, constant $c_p = 410.94444 \text{ J kg}^{-1} \text{ K}^{-1}$, constant $k = 9.870588 \text{ W m}^{-1} \text{ K}^{-1}$ and no latent heat. Thus $a_0 = k/(\rho c_p) = 2.845886 \times 10^{-6} \text{ m}^2/\text{s}$. Its finite-domain mesh contains 178992 cells ($113 \times 33 \times 48$), with near-source spacings of $8 \times 6 \times 2 \mu\text{m}$. Domain extents follow the production geometry, but the adiabatic transverse and bottom boundaries truncate the semi-infinite analytical referent.

A constant-property moving-Gaussian reference is constructed from the semi-infinite heat kernel, in the moving-source formulation [3]. With $Q = \eta P$, convolving the surface Gaussian with the Neumann kernel gives

$$T - T_0 = \frac{Q}{\rho c_p \pi \sqrt{\pi a_0}} \int_0^\infty \frac{\exp \left[-\frac{(\xi + v\tau)^2 + y^2}{w^2/2 + 4a_0\tau} - \frac{z^2}{4a_0\tau} \right]}{\sqrt{\tau}(w^2/2 + 4a_0\tau)} d\tau. \quad (\text{A.4})$$

The archived implementation removes the surface singularity by $\tau = (w/v)s^2$ and uses adaptive quadrature. Solidus geometry is extracted at $1290 \text{ }^\circ\text{C}$, with uniform-core face-average source recovery

for the numerical surface. Width/depth use the monotone constant-property maximum-temperature envelope; they do not replace the general production passage observer. The associated recipes and archived reports are identified in the numerical reproduction notes. The quoted differences combine discretization, reconstruction and finite-domain effects, and do not bound nonlinear phase-change or physical-model error.

Appendix A.6. Calibration record and retained sensitivities

The calibration bracket $\eta \in [0.2, 0.4]$ has retained endpoint lengths of 234.57 and 511.68 μm . Subsequent retained trial pairs $(\eta, L/\mu\text{m})$ are (0.2898051, 359.96) and (0.2853148, 354.22), followed by the final (0.2890548519, 359.16). They establish a sampled local bracket, not global identifiability. The local secant sensitivity is approximately 1276.79 μm per unit η . Dividing B's 21.26 μm expanded uncertainty by this slope gives a scale $\Delta\eta = 0.01665$; it is not a posterior interval and is not propagated into the geometry comparisons.

Table A.5: Retained calibrated axial sensitivities in μm , calculated as fine minus baseline. Only the near-source axial spacing changes from 4 to 2 μm ; transverse and depth spacings remain 3 and 1 μm . These contrasts do not bound every discretization component.

Condition	ΔL	ΔW	ΔD
B	+0.361	−0.009	+0.041
C	+0.199	−0.029	+0.045

For uncertainty-scale reporting, $\delta/U = (Q_{\text{model}} - \overline{Q}_{\text{source}})/U_{\text{Lane}}$. The separate U values are not specimen standard deviations. Imaging and microscopy have different operators and aggregation, so the geometry triplet is not fitted to a joint likelihood. All finite contrasts use full-precision outputs before rounding for display.

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